Note 4.2 Matrix Representation of Linear Transformations

Theorem and Corollary

Theorem 4.2.1

If L is a linear transformation mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, then there is an $m \times n$ matrix A such that

$$L(\mathbf{x}) = A\mathbf{x}$$

Then, the jth column vector of A is given by

$$\mathbf{a}_j = L(\mathbf{e}_j), j = 1, 2, \cdots, n$$

Proof

Since $L(\mathbf{x}) = A\mathbf{x}$, then

$$L(\mathbf{x}) = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

let $\mathbf{x} = \mathbf{e}_j$, then $x_j = 1$ and other scalars are zero. That is,

$$L(\mathbf{e}_j) = \mathbf{a}_j$$

In fact, when representing the basis of $\mathbf{x}$ as the standard unit basis, the proof can also be done.

The theorem tells us how to construct the matrix A corresponding to a particular linear transformation of L. We refer to A as the standard matrix representation of L.

Theorem 4.2.2 Matrix Representation Theorem

If $E = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}, F = \{\mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_m\}$ are ordered bases for vector space V and W, then corresponding to each linear transformation $L : V \to W$, there is an $m \times n$ matrix A such that

$$[L(\mathbf{v})]_F = A[\mathbf{v}]_E$$

for each $\mathbf{v} \in V$. It is similar to Note 3.5 Change of Basis > Transition Matrix.
A is the matrix representing L relative to the ordered bases E and F. And

$$\mathbf{a}_j = [L(\mathbf{v}_j)]_F, j = 1, 2, \cdots, n$$

$$\begin{tikzcd}
\mathbf{v} \in V \arrow[r, "L = L_A"] \arrow[d, leftrightarrow] & \mathbf{w} = L(\mathbf{v}) \in W \arrow[d, leftrightarrow] \\
\mathbf{x} = [\mathbf{v}]_E \in R^n \arrow[r, "A"] & A\mathbf{x} = [\mathbf{w}]_F \in R^m
\end{tikzcd}$$

Figure 4.2.2.

Proof

Suppose

$$\mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n$$

and

$$L(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \cdots + a_{mj}\mathbf{w}_m, 1 \le j \le n$$

Here is

$$\mathbf{a}_j = [L(\mathbf{v}_j)]_F$$

Thus

$$\begin{aligned} L(\mathbf{v}) &= L(x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n) \\ &= (x_1a_{11} + x_2a_{12} + \cdots + x_na_{1n})\mathbf{w}_1 + \cdots + (x_1a_{m1} + \cdots + x_na_{mn})\mathbf{w}_n \\ &= \sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}x_j\right)\mathbf{w}_i \end{aligned}$$

Then let

$$y_i = \sum_{j=1}^{n} a_{ij}x_j$$

Thus

$$\mathbf{y} = (y_1, y_2, \cdots, y_m)^T = A\mathbf{x}$$

Strategy

To find the matrix representation A for a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ with respect to the ordered bases $E = \{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\}$ and $F = \{\mathbf{b}_1, \cdots, \mathbf{b}_m\}$, we need to represent each vector $L(\mathbf{u}_j)$ as a linear combination of $\mathbf{b}_1, \cdots, \mathbf{b}_m$.

Theorem 4.2.3

Let $E = \{\mathbf{u}_1, \cdots, \mathbf{u}_n\}$ and $F = \{\mathbf{b}_1, \cdots, \mathbf{b}_n\}$ be ordered bases for $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $L : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and A is the matrix representing L with respect to E and F, then

$$\mathbf{a}_j = B^{-1}L(\mathbf{u}_j), j = 1, \cdots, n$$

where $B = (\mathbf{b}_1, \cdots, \mathbf{b}_m)$.

Proof

If A is representing L with respect to E and F, then,

$$L(\mathbf{u}_j) = a_{1j}\mathbf{b}_1 + \cdots + \mathbf{a}_{mj}\mathbf{b}_m = B\mathbf{a}_j$$

Clearly B is nonsingular whose column vectors are the basis for $\mathbb{R}^m$. Thus

$$\mathbf{a}_j = B^{-1}L(\mathbf{u}_j), j = 1, \cdots, n$$

Corollary 4.2.4

If A is the matrix representing the linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ with respect to the bases

$$E = \{\mathbf{u}_1, \cdots, \mathbf{u}_n\}, F = \{\mathbf{b}_1, \cdots, \mathbf{b}_m\}$$

then the reduced row echelon form of $(\mathbf{b}_1, \cdots, \mathbf{b}_m | L(\mathbf{u}_1), \cdots, L(\mathbf{u}_n))$ is $(I|A)$.

It can be obtained by multiplying B^{-1} of the both sides. (The left side can be seen as B).

Application I Computer Graphics and Animation

For the picture in the plane stored in the computer, it is stored as a set of vertices. Then the vertices can be plotted and connected by lines to produce the picture. If there are n vertices, they are stored in a $2 \times n$ matrix. The x-coordinates of the vertices are stored in the first row and the y-coordinates of the vertices are in the second.
We can transform a figure by changing the positions of the vertices and then drawing the figure. And the primary transformations are as follows.

- Dilations and contractions
  The linear operator of the form

  $$L(\mathbf{x}) = c\mathbf{x}$$

  is dilation if $c > 1$ and contraction if $0 < c < 1$.
- Reflection about an axis.
  If L_x is a transformation that reflects a vector $\mathbf{x}$ about x-axis, then

  $$L_x(\mathbf{e}_1) = \mathbf{e}_1, L_x(\mathbf{e}_2) = -\mathbf{e}_2$$

Thus the matrix A is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Similarly, the matrix representation for L_y is $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

- Rotations
  The matrix A in the linear operator $L(\mathbf{x}) = A\mathbf{x}$ is

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

  That is rotating the vector by θ in the counterclockwise direction.
- Translation
  Translation is a transformation by a vector $\mathbf{a}$ in the form that

$$L(\mathbf{x}) = \mathbf{x} + \mathbf{a}$$

  Note that if $\mathbf{a} \neq \mathbf{0}$, then L is not a linear transformation.
  So L can not be represented by a 2×2 matrix.

Homogeneous Coordinates

The homogeneous coordinate system is formed by equating each vector in $\mathbb{R}^2$ with a vector in $\mathbb{R}^3$ having the same first two coordinates and having 1 as its third coordinate:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leftrightarrow \begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix}$$

When discussing the upper linear transformation, take the 2×2 matrix representation and augment it by attaching it the third row and third column of the 3×3 identity matrix. For example, the dilation matrix

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

and corresponding matrix is

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

such that

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3x_1 \\ 3x_2 \\ 1 \end{pmatrix}$$

So we transform the pair (x_1, x_2) and just ignore the third coordinate.

For a translation, we can find a matrix representation for L with respect to homogeneous coordinate system. We simply take the 3×3 identity matrix and replace the first two entries of its third column with the entries of $\mathbf{a}$. For example, $\mathbf{a} = (6, 2)^T$. Then

$$A\mathbf{x} = \begin{pmatrix} 1 & 0 & 6 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} = \begin{pmatrix} x_1 + 6 \\ x_2 + 2 \\ 1 \end{pmatrix}$$